Problem 1: Constraints Elimination (0.5 pt)

Solve the following non-lineal optimization problem in $\mathbb{R}^3$ with lineal equality constraints by constraint elimination:

$$\text{minimize } f(x) = \exp(\mathbf{1}^T x) + \exp(-\mathbf{1}^T x)$$
$$\text{subject to: } \begin{pmatrix} 1 & -2 & 0 \\ 2 & 1 & 0 \end{pmatrix} x = \begin{pmatrix} 0 \\ 5 \end{pmatrix}$$

$\Rightarrow \min\limits_{z} f(Fz + x_0) = \hat{f}(z)$

$F \in \mathbb{R}^{3\times 1}$: basis of $N(A)$

$x_0 \in \mathbb{R}^3$: any x satisfying $Ax = b$.

Hint: use Gaussian elimination or QR decomposition. You are allowed to use a computer algebra system to verify your computations.

$N(A) = \begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} = Fz \Rightarrow \boxed{F = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}}$

Gaussian elimination

$\left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 2 & 1 & 0 & 5 \end{array}\right) \Rightarrow \left(\begin{array}{ccc|c} 1 & -2 & 0 & 0 \\ 0 & 5 & 0 & 5 \end{array}\right) \Rightarrow \begin{array}{l} x_1 = 2 \\ x_2 = 1 \end{array}$, chose $x_3 = 0$ $\Rightarrow \boxed{x_0 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}}$

$\Rightarrow \min\limits_{z} f\left(\begin{bmatrix} 0 \\ 0 \\ z \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}\right) = e^{3+z} + e^{-3-z}$. $\hat{f}'(z) = e^{3+z} - e^{-3-z} \overset{!}{=} 0 \Rightarrow 3+z = -3-z \Rightarrow \boxed{z^* = -3}$

$\Rightarrow \boxed{x^*} = Fz + x_0 = \begin{bmatrix} 0 \\ 0 \\ -3 \end{bmatrix} + \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} = \boxed{\begin{bmatrix} 2 \\ 1 \\ -3 \end{bmatrix}}$

Problem 2: Solving the KKT system (0.5 pt)

Solve the following convex quadratic problem in $\mathbb{R}^3$ with linear equality constraints:

$$\min_{x\in\mathbb{R}^3} \quad f(x) = (-3 \ \ -2 \ \ 7)x + \tfrac{1}{2}x^\top \begin{pmatrix} 6 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 6 \end{pmatrix} x$$

$$\text{s.t.} \quad \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix} x = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

Hint: use Gaussian elimination or QR decomposition to solve the resulting KKT system.

Solution 2. We write the problem in the standard quadratic form

$$\min_{x\in\mathbb{R}^3} \ \tfrac{1}{2}x^\top Qx + c^\top x \quad \text{s.t.} \quad Cx = d,$$

with

$$Q = \begin{pmatrix} 6 & 0 & 2 \\ 0 & 4 & 0 \\ 2 & 0 & 6 \end{pmatrix}, \qquad c = \begin{pmatrix} -3 \\ -2 \\ 7 \end{pmatrix}, \qquad C = \begin{pmatrix} 1 & 1 & 2 \\ 0 & 1 & 1 \end{pmatrix}, \qquad d = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

Introducing Lagrange multipliers $\lambda \in \mathbb{R}^2$ for the equality constraints, the KKT conditions are

$$\begin{aligned} Qx + c + C^\top\lambda &= 0, \\ Cx - d &= 0. \end{aligned}$$

Equivalently, we obtain the linear KKT system

$$\begin{pmatrix} 6 & 0 & 2 & 1 & 0 \\ 0 & 4 & 0 & 1 & 1 \\ 2 & 0 & 6 & 2 & 1 \\ 1 & 1 & 2 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \\ \lambda_1 \\ \lambda_2 \end{pmatrix} = \begin{pmatrix} 3 \\ 2 \\ -7 \\ 0 \\ 2 \end{pmatrix},$$

where the right-hand side comes from $Qx + C^\top\lambda = -c$ and $Cx = d$.

We now solve this linear system using Gaussian elimination. The augmented matrix is

$$\left[\begin{array}{ccccc|c} 6 & 0 & 2 & 1 & 0 & 3 \\ 0 & 4 & 0 & 1 & 1 & 2 \\ 2 & 0 & 6 & 2 & 1 & -7 \\ 1 & 1 & 2 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 & 2 \end{array}\right].$$

Applying Gaussian elimination (row operations), we reduce it to row-echelon form and then to reduced row-echelon form. The resulting matrix is

$$\left[\begin{array}{ccccc|c} 1 & 0 & 0 & 0 & 0 & -1 \\ 0 & 1 & 0 & 0 & 0 & 3 \\ 0 & 0 & 1 & 0 & 0 & -1 \\ 0 & 0 & 0 & 1 & 0 & 11 \\ 0 & 0 & 0 & 0 & 1 & -21 \end{array}\right],$$

which is the reduced row-echelon form of the KKT system.

From this we read off the unique solution

$$x_1^\star = -1, \qquad x_2^\star = 3, \qquad x_3^\star = -1,$$

and the corresponding Lagrange multipliers

$$\lambda_1^\star = 11, \qquad \lambda_2^\star = -21.$$

One can verify that the constraints are satisfied:

$$\begin{gathered} x_1^\star + x_2^\star + 2x_3^\star = -1 + 3 + 2(-1) = 0, \\ x_2^\star + x_3^\star = 3 + (-1) = 2. \end{gathered}$$

Since Q is positive definite, the quadratic objective is strictly convex and the KKT conditions are sufficient for optimality. Therefore $x^\star = (-1, 3, -1)^\top$ is the unique minimizer of the problem.

The minimal objective value is

$$f(x^\star) = (-3 \ \ -2 \ \ 7)x^\star + \tfrac{1}{2}(x^\star)^\top Q x^\star = 16.$$